Hiten Bahrani

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SUMMARY

Data Analyst and MS Information Systems candidate with hands-on experience using Python and SQL to prepare, analyze, and report on data. Skilled in building ETL workflows and Tableau and Power BI dashboards that turn operational data into clear insights for technical and business stakeholders. Brings a strong foundation in statistics, data visualization, and communicating findings clearly.

TECHNICAL SKILLS

- **Data Analysis**: Python, SQL, pandas, NumPy, Data preparation, ETL, Feature engineering
- **Business Intelligence & Visualization**: Tableau, Power BI, Matplotlib, Seaborn, Dashboard Development
- **Statistics & Analytics**: hypothesis testing, A/B testing fundamentals, regression analysis, model evaluation
- **Databases & Cloud**: MySQL, PostgreSQL, AWS (EC2, S3)
- **Tools**: Git, GitHub, VSCode, JIRA
- **Additional Technical Skills**: Scikit-learn, SHAP, FastAPI, REST APIs, React, RAG, Prompt Engineering

EXPERIENCE

University of Texas at Arlington | ***Graduate Teaching Assistant*** **Jan 2026 - Present**

- Evaluated high-volume Python programming assignments using consistent grading rubrics, with close attention to data structures, algorithms, and edge-case correctness.
- Applied structured partial-credit criteria to ensure accurate, fair, and efficient evaluation across student submissions.
- Tracked recurring error patterns across cohorts, including off-by-one errors, mutation bugs, and performance issues, and shared key trends with the course instructor.

ParticleBlack | ***Data Analyst*** **Jun 2024 - Dec 2024**

- Built Python and SQL ETL workflows using joins, aggregations, and windowed metrics to prepare more than 50,000 conversational interaction logs for analysis and modeling.
- Analyzed conversation metadata, including turn count, intent transitions, and session length, to identify factors associated with query resolution and support chatbot-routing decisions.
- Developed Tableau and Power BI dashboards to monitor chatbot resolution rates and surface performance trends and potential model-drift signals for stakeholders.

Mohan Group | ***Data Analyst*** **Dec 2022 - Jun 2024**

- Integrated 4 record types (property, lead, booking, and payment) using SQL and Python; built Power BI dashboards to track inventory, sales velocity, conversion rates, and overdue collections.
- Analyzed 2 conversion metrics (lead-to-visit and visit-to-booking) by marketing channel and response time to identify drop-off points and guide sales follow-up.
- Evaluated and ranked development sites using ArcGIS, demographics, competitor pricing, transport access, and historical sales to support management's site-selection decisions
- Mapped buyer and lead locations to define customer catchments, assess overlap between developments, and identify gaps in market coverage for expansion and targeted marketing.
- Documented data sources, quality checks, and site-scoring assumptions to support reproducible reporting and analysis.

YHills | ***Data Scientist and Machine Learning Trainee*** **Jun 2022 - Sep 2022**

- Prepared and analyzed a dataset of 10,000+ taxi trip records to examine how distance, time of day, and traffic conditions influenced fare estimates.
- Built and evaluated linear regression and random forest models to compare predictive performance across fare-prediction approaches.
- Set up AWS EC2 and S3 resources to support repeatable dataset storage and model-training workflows outside the local environment.

PROJECTS

ML Decision Support with Explainability **May 2025 - Jun 2025**

- Built a classification-based decision-support tool with rule-based policy thresholds that routed low-confidence predictions for human review.
- Used SHAP to examine the key factors influencing individual predictions and review subgroup performance for potential fairness gaps.
- Created a Streamlit dashboard that allowed stakeholders to explore prediction confidence, threshold trade-offs, and fairness diagnostics.

LLM-Powered Research Assistant | ***Nova Nexus AI*** **Nov 2025 - Feb 2026**

- Built a full-stack research assistant that combined LLM responses with optional web retrieval for time-sensitive queries.
- Developed a FastAPI backend to route queries, retrieve relevant sources, and provide grounded context to the model.
- Built a React frontend with conversational memory, structured responses, and inline source citations; deployed the application on Vercel and Render.

EDUCATION

University of Texas at Arlington | ***Master of Science, Information Systems*** **Arlington, TX Jan 2025 - Dec 2026**

Thadomal Shahani Engineering College | ***Bachelor of Technology, Information Technology*** **Mumbai, India Sep 2020 - Jun 2024**

CERTIFICATIONS

- Introduction to Data Management: Meta (2026)
- Data Science Methodology: IBM (2026)
- Introduction to AI : Coursera (2025)
- Google Data Analytics Professional Certificate: Google (2024)